HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY AND EDUCATION

FACULTY OF ADVANCED TRAINING

COURSE REPORT



DIGITAL SIGNAL PROCESSING

PREPARED BY: GROUP 7

1. Trương Thuận Phát 23146026
2. Lê Bình Nguyên 23146019
3. Trương Minh Quang 23146032
4. Lê Bá Hoàng Anh 22146003

COURSE CODE: DIPR337529E

SUPERVISOR: Dr. Bùi Hà Đức

Ho Chi Minh City, October 22, 2026

Contents

CHAPTER 1. INTRODUCTION 1
1.1. Problem statement and objectives 1
1.1.1. Bluetooth-based distance estimation problem 1
1.1.2. Project objectives and requirements 1
1.2. Theoretical background 1
1.2.1. Bluetooth Low Energy, Advertising, and RSSI 1
1.2.2. Sources of RSSI noise 2
1.2.3. Log-distance path-loss model 3
1.3. System concept and architecture 4
1.3.1. System architecture block diagram 4
1.3.2. Hardware and software used 4
CHAPTER 2. METHODOLOGY 5
2.1. Data collection and determination of distance-model parameters 5
2.1.1. Experimental setup and CSV data format 5
2.1.2. Preprocessing and spectral analysis using Welch's method 6
2.1.3. Log-distance regression, 95% confidence intervals, and the quadratic model 7
2.2. Noise filter design 9
2.3. Real-time algorithm and evaluation methods 10
2.3.1. Distance formula and algorithm flowchart 10
2.3.2. Leave-one-out validation, RMSE, and the proportion of samples within ±0.5 m and ±1 m 11
2.3.3. Converting the ±1 m tolerance to dB at each distance 12
CHAPTER 3. RESULTS 14
3.1. Survey and calibration results 14
3.2. Noise filtering results 15
3.3. Accuracy evaluation and conclusions 17
References 17
Source code 17

CHAPTER 1. INTRODUCTION

1.1. Problem statement and objectives

1.1.1. Bluetooth-based distance estimation problem

Distance estimation based on Bluetooth signals is essentially the measurement of the received signal strength indicator (RSSI) at the receiving device, followed by applying a radio-wave power-attenuation model to infer the physical distance. The biggest challenge is that 2.4 GHz signals are affected in complex ways by the propagation environment, and are easily distorted and corrupted by multipath, obstacles, co-channel interference, and antenna orientation.

1.1.2. Project objectives and requirements

- Accuracy: The system achieves a distance-estimation error within ±1 m in an indoor environment.
- Operability: RSSI sequences are collected and processed in real time.
- Stability: The software must apply filters (Median & Kalman Filter) to suppress impulsive noise and reduce random signal fluctuations.

1.2. Theoretical background

1.2.1. Bluetooth Low Energy, Advertising, and RSSI

BLE operates in the 2.4 GHz ISM band, divided into 40 channels of 2 MHz each. Of these, 37 channels are used for connection data transmission, while the remaining 3 channels are 37, 38, and 39, which are reserved for Advertising. The scanning device listens for advertising packets without needing to establish a connection, while measuring the energy at the antenna and converting it into an RSSI value (in dBm). An RSSI that is negative and closer to 0 indicates a stronger signal.

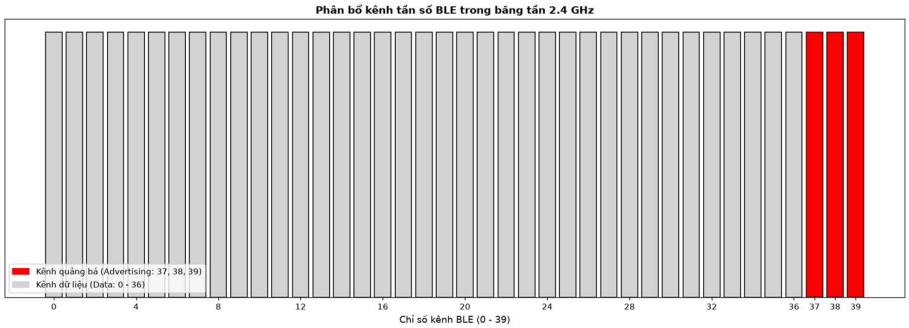


Figure 1.1: BLE frequency channel allocation in the 2.4 GHz band

1.2.2. Sources of RSSI noise

Computing distance from RSSI is difficult because of the following main noise sources:

- Multipath: Waves reflect off walls, ceilings, floors, and furniture, so the component waves interfere with each other and cause abrupt changes in RSSI.

- Fading: Shadowing by large obstacles and the movement of people/objects cause the signal to attenuate quickly or slowly.

- Device orientation: Rotating the phone between landscape and portrait changes the radiated power received.

- Co-channel interference: Signal contention with Wi-Fi networks and other Bluetooth devices operating in the same 2.4 GHz band.

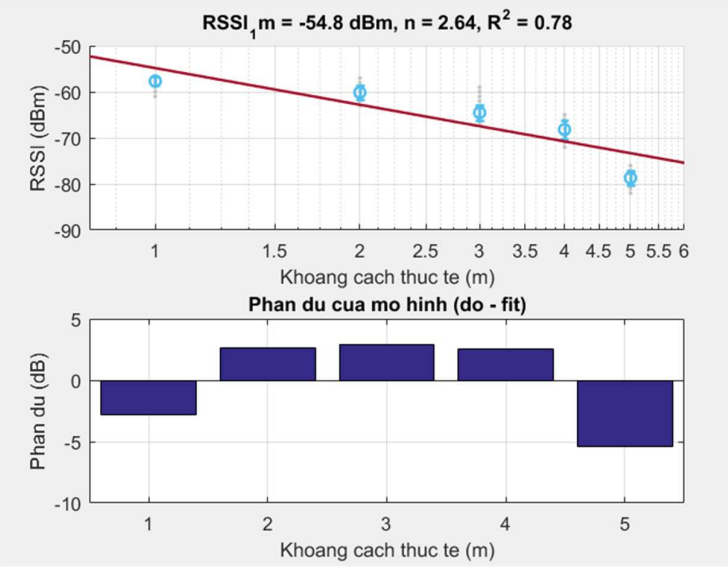


Figure 1.2: Chart showing deviation from the true distance

Theoretical basis for applying filters:

To understand why the system must use filters, we collected samples and analyzed the signal spectrum:

- Chart 1.2 (Signal noise characteristics): A test collected 100 samples at distance marks d = 1 m, 2 m, 3 m, 4 m, 5 m. The results show that as the distance (d) increases, the fluctuation around the mean grows larger. This demonstrates that measurement error increases in proportion to distance.

- Chart 1.3 (Spectral Analysis): By converting the raw signal to the frequency domain (f), the signal can be seen to consist of a low-frequency band (representing actual, slow human movement) and a high-frequency band that oscillates chaotically (representing environmental noise and wave interference).

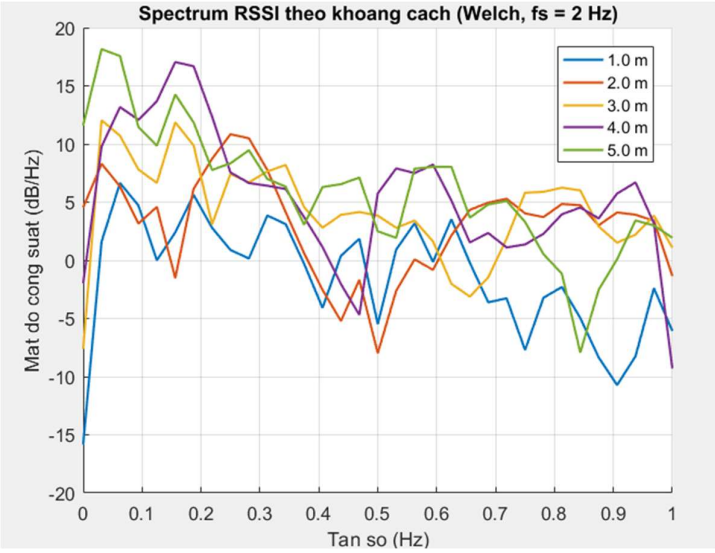


Figure 1.3: Frequency spectrum analysis chart of the RSSI signal

Based on this spectral chart, we decided to implement a Median filter combined with a Kalman Filter to predict and compensate for signal delay, helping to optimize the value.

1.2.3. Log-distance path-loss model

The most common model for converting RSSI into distance (d) is the log-distance model. The core equation applied in practice is:

$$d = 10^{\frac{(\mathrm{RSSI}_{1M} - \mathrm{RSSI})}{(10 \cdot \mathrm{n})}}; [1]$$

Where RSSI_{1M} is the RSSI value at a reference distance of 1 meter, and n is the path-loss exponent of the environment.

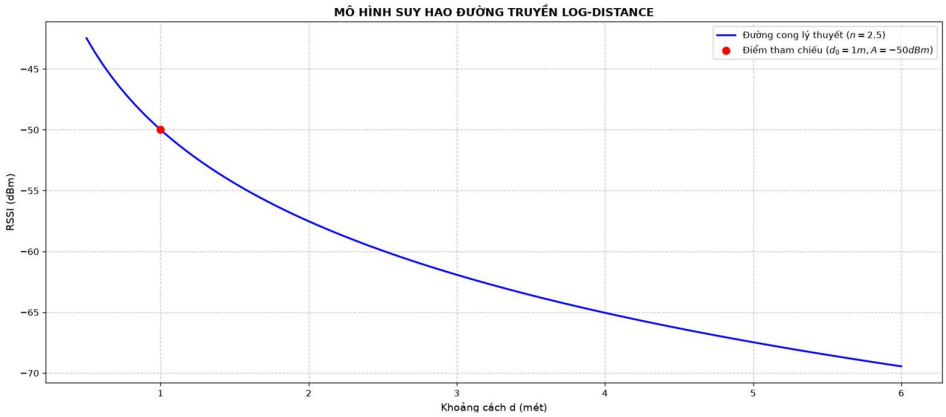


Figure 1.4: Plot of the theoretical log-distance path-loss curve

1.3. System concept and architecture

1.3.1. System architecture block diagram

The positioning system is built on Bluetooth Low Energy (BLE) technology, consisting of Device A and Device B, which transmit signals, and Device C which collects the RSSI signal strength from the two devices. Then, Device C sends the RSSI_A data, RSSI_B, and its own RSSI to the computer via Bluetooth web. The computer receives and processes the data and applies a suitable algorithm to estimate the distance or determine the position of the phone. Finally, the results are displayed on the computer interface. The system enables position tracking based on BLE signal strength.

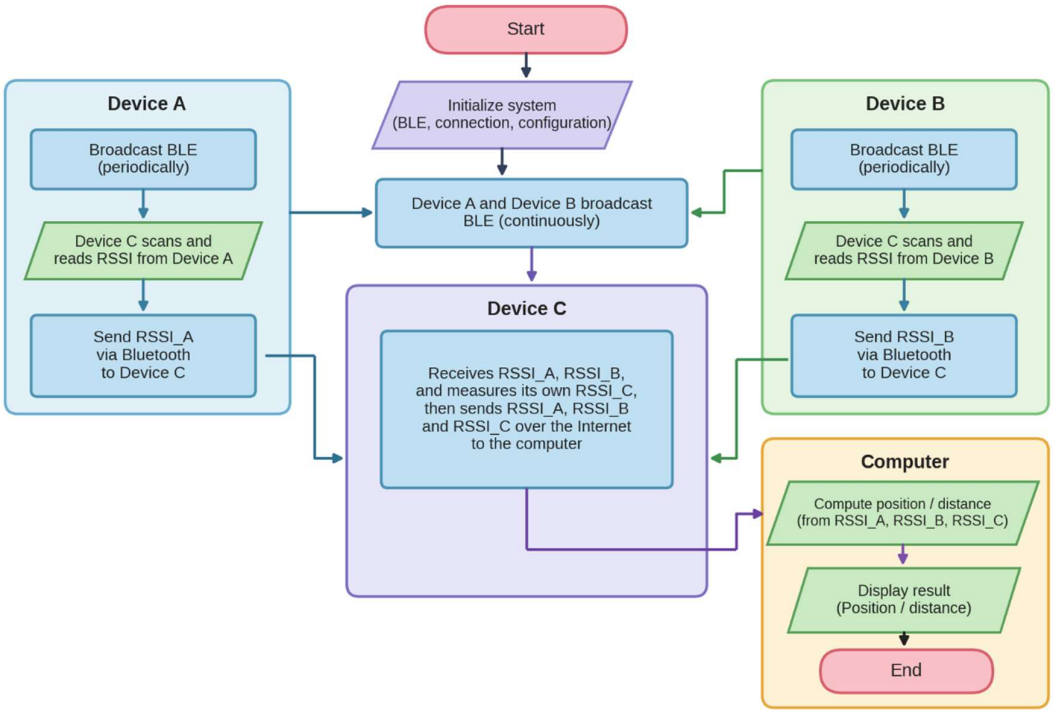


Figure 1.4: System block diagram

1.3.2. Hardware and software used

- Hardware: 1 smartphone (realme8) whose position can be changed, acting as a transceiver, 2 smartphones act as Anchors continuously broadcasting BLE. A Windows laptop acts as both an Anchor and a receiving device.

- Programming language: Python.

- Libraries: bleak (real-time BLE signal scanning), scipy (trilateration solver), matplotlib (plotting).

CHAPTER 2. METHODOLOGY

2.1. Data collection and determination of distance-model parameters

2.1.1. Experimental setup and CSV data format

The measurement system consists of a computer (PC) placed at the origin, DeviceB at (4; 0) m on the x-axis, DeviceA at (0; 4) m on the y-axis, and DeviceC as the device to be located (Figure 2.1). There are three RSSI measurement paths: $RSSI_A$ and $RSSI_B$ measured by DeviceC from the BLE signals of DeviceA and DeviceB (via Web Bluetooth on Chrome, sent to the PC over HTTP); $RSSI_C$ measured by the PC from the advertisements of DeviceC using the bleak library. Each path has different transmit–receive hardware and is therefore calibrated separately, with its own parameter pair ($RSSI_{1M}$, n).

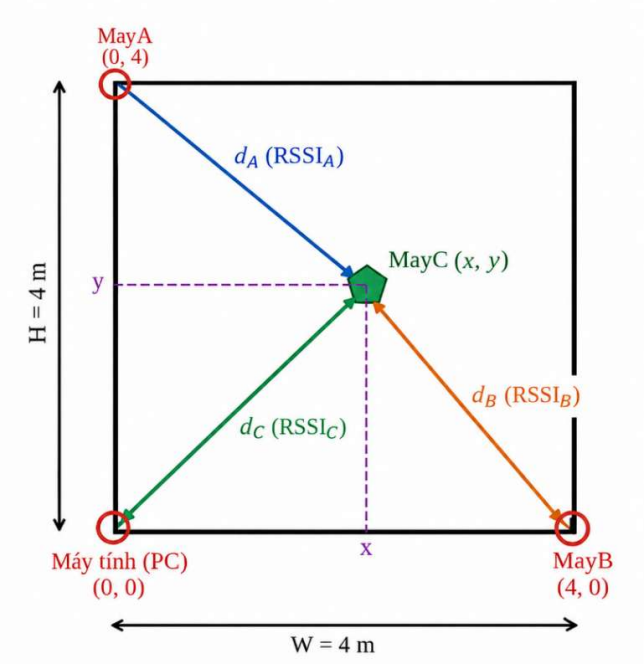


Figure 2.1: Measurement setup: three RSSI paths (d_A, d_B, d_C) and the coordinates to be determined (x, y).

For each path, the transmit–receive pair is placed d = 1, 2, 3, 4, 5 m apart; the orientation and height of the devices are kept fixed across levels; each level is recorded continuously for about 60 s (about 200 samples in total).

Samples arrive on an event basis and are therefore not evenly spaced: the average sampling rate is 3.6–4.6 Hz (DeviceA), 4.9–5.6 Hz (DeviceB), and 1.5–2.3 Hz (DeviceC). Because RSSI is an integer, there is a quantization error of 1 dB. Statistics for each level are summarized in Table 2.1; the time series are shown in Figure 2.2. The mean of DeviceA (−74.4 → −84.7 dBm) and DeviceC (−58.0 → −79.1 dBm) decreases with distance, but DeviceB is not monotonic: −83.1; −88.3; −82.7 dBm at 3; 4; 5 m.

Table 2.1. Raw RSSI statistics by path and distance level.

Path	d (m)	N	f_s (Hz)	Mean (dBm)	σ (dB)	Min / Max (dBm)
DeviceA	1	271	4.51	−74.36	1.18	−79 / −72
DeviceA	2	277	4.61	−75.86	2.41	−82 / −71
DeviceA	3	218	3.62	−81.97	4.88	−95 / −73
DeviceA	4	239	3.98	−83.65	2.00	−89 / −80
DeviceA	5	235	3.90	−84.71	3.26	−94 / −79
DeviceB	1	321	5.34	−77.31	3.29	−85 / −72
DeviceB	2	334	5.57	−77.71	3.43	−93 / −72
DeviceB	3	302	5.02	−83.14	1.25	−88 / −78
DeviceB	4	294	4.89	−88.26	3.46	−96 / −82
DeviceB	5	293	4.87	−82.74	1.67	−88 / −79
DeviceC	1	138	2.33	−57.99	1.29	−61 / −56
DeviceC	2	136	2.26	−60.14	1.94	−66 / −57
DeviceC	3	126	2.09	−64.32	2.05	−68 / −59
DeviceC	4	120	2.02	−68.21	2.42	−75 / −64
DeviceC	5	88	1.48	−79.14	2.72	−91 / −74

2.1.2. Preprocessing and spectral analysis using Welch's method

Because the samples are not evenly spaced, each series is linearly interpolated onto a grid before computing the power spectral density (PSD) with Welch's method: Hann window, segments of at most 64 samples, 50% overlap, with the mean of each segment subtracted; the frequency resolution is about 0.03–0.08 Hz. No samples were removed from the calibration dataset.



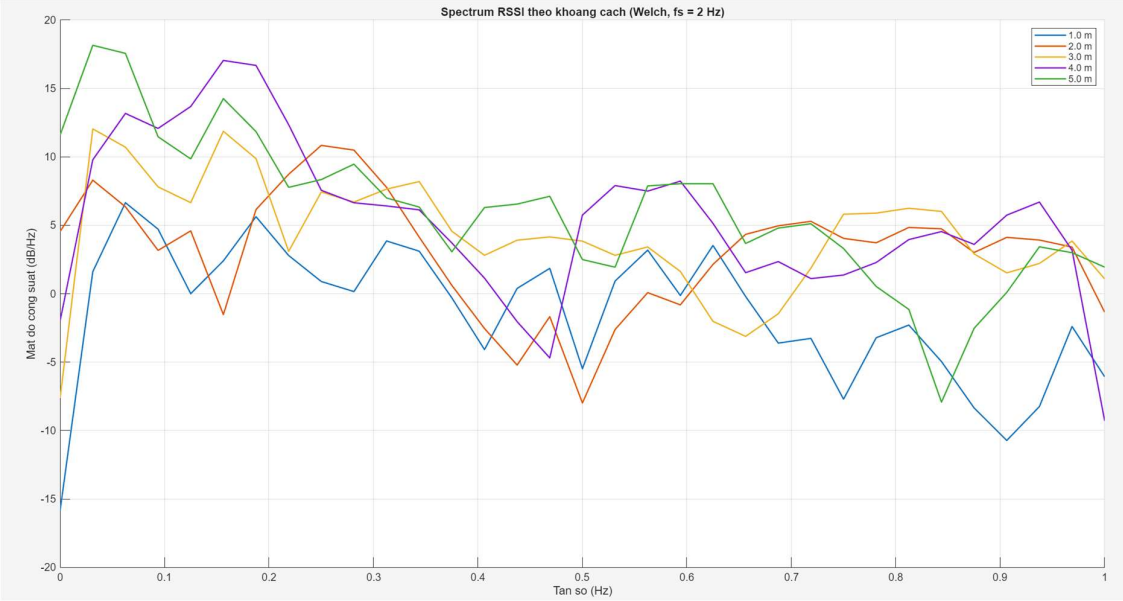


Figure 2.2: Welch power spectral density of RSSI for each path and distance level.

Figure 2.3 shows that the spectra of DeviceA and DeviceB are nearly flat, i.e., the noise is nearly white: 61–79% of the power lies at frequencies $f \geq 0.5$ Hz and only 2–9% at $f \leq 0.1$ Hz; the lag-1 autocorrelation coefficient lies between −0.24 and 0.19. For DeviceC, the spectrum is skewed toward low frequencies (only 6–26% of the power at $f \geq 0.5$ Hz, 12–32% at $f \leq 0.1$ Hz; lag-1 autocorrelation from 0.27 to 0.51). Two consequences for filter design: (i) most of the noise energy lies outside the band of slow human movement, so a low-pass filter with a cutoff frequency much lower than 0.5 Hz can be used; (ii) for the DeviceC path, the noise is correlated, so filtering reduces noise less effectively than in the white-noise case.

2.1.3. Log-distance regression, 95% confidence intervals, and the quadratic model

The log-distance model relates RSSI to distance d (m):

$$RSSI(d) = RSSI_{1M} - 10.n.\log_{10}(d) \quad (2.1)$$

The two parameters are estimated by ordinary least squares (OLS) of RSSI against $\log_{10}(d)$ over all raw samples of each path. Because consecutive samples are correlated, the 95% confidence interval (CI) is computed by block bootstrap (blocks of 10 samples, resampled within each level, 1000 iterations); to reflect the uncertainty arising from having only 5 distance levels, the CI is also computed from a regression on the 5 level means (t-distribution, 3 degrees of freedom).

When the log-distance model does not fit, a quadratic model in d is used:

$$RSSI(d) = a + b.d + c.d^2 \quad (2.2)$$

The two models are compared using:

R^2, RMSE (dB) and the AIC criterion: ($\Delta AIC = AIC_{\log} - AIC_{\text{quad}}$) on all samples; the quadratic model is chosen only when it is clearly better and monotonically decreasing over 1–5 m, because only then can RSSI be inverted into distance.

Table 2.2. Regression results of RSSI versus $\log_{10}$(d) and the quadratic model.

Parameter	**DeviceA**	**DeviceB**	**DeviceC**
$RSSI_{1M}$ (dBm)	−73.3	−76.5	−55.5
95% CI (bootstrap)	[−73.5; −73.1]	[−76.7; −76.2]	[−55.8; −55.3]
95% CI (5 levels, t)	[−77.8; −68.9]	[−85.6; −67.5]	[−68.2; −41.8]
n	1.62	1.28	2.43
95% CI (bootstrap)	[1.57; 1.67]	[1.23; 1.33]	[2.34; 2.49]
95% CI (5 levels, t)	[0.70; 2.55]	[−0.60; 3.15]	[−0.09; 5.36]
R^2 (log-distance)	0.62	0.42	0.69
RMSE log-distance (dB)	3.20	3.75	4.00
R^2 (quadratic)	0.64	0.47	0.90
RMSE quadratic (dB)	3.10	3.59	2.24
Extremum of quadratic at d (m)	7.3	4.3	0.9
$AIC_{log} - AIC_{quad}$	74	140	703

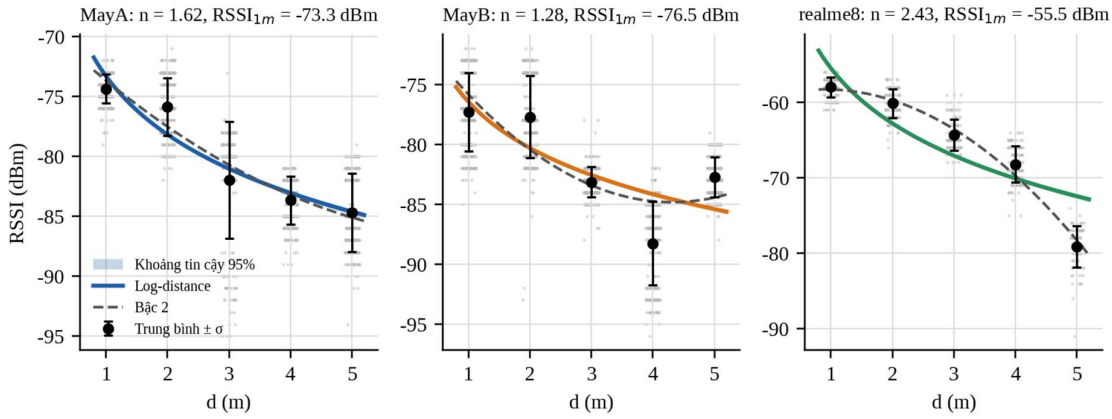


Figure 2.3: RSSI regression versus distance: mean ± σ at each level, log-distance curve with 95% CI (bootstrap), and quadratic curve.

DeviceA has an RSSI that decreases steadily with distance, so log-distance fits reasonably well (n = 1.63, R^2 = 0.62); the quadratic model only raises R^2 slightly to 0.64 and has its extremum at d ≈ 7.3 m, outside the measured range, so log-distance is retained for this path.

For DeviceB, RSSI is not monotonic (weakest at 4 m), so log-distance gives only $R^2 = 0.42$ and $n = 1.28$; the quadratic model has a large ΔAIC (140) but has a minimum at $d \approx 4.3$ m, i.e., it is not monotonic over 1–5 m and cannot be inverted at $d > 4.3$ m.

For DeviceC, the quadratic model is clearly superior (R^2 0.90 versus 0.69; RMSE 2.24 dB versus 4.00 dB; $\Delta AIC \approx 703$) and is monotonically decreasing over 1–5 m, because RSSI drops faster than log-distance predicts at 5 m (−79.1 dBm). The bootstrap CI is very narrow because it reflects only measurement noise; the CI from the 5 levels is much wider (n of DeviceA lies in [0.70; 2.55]) and is a more realistic measure of the certainty of the model.

2.2. Noise filter design

The RSSI sequence is filtered through two cascaded stages. Stage 1 is a moving average over the N = 9 most recent samples (when fewer than 9 samples are available, the average of the available samples is used), which reduces the influence of isolated outliers. Stage 2 is a one-dimensional Kalman filter with a random-walk state model: $x_k = x_{k-1} + w_k$, $z_k = x_k + v_k$, with $w \sim N(0, Q)$ and $v \sim N(0, R)$. Each new sample z_k is updated as follows:

$$P_k^- = P_{k+1} + Q;\ K_k = P_k^-\ /\ (P_k^- + R) \tag{2.3}$$

$$x_k = x_{k-1} + K_k \cdot (z_k - x_{k-1});\ P_k = (1 - K_k) \cdot P_k^- \tag{2.4}$$

The filter is initialized with the first sample, with Q = 0.03 and R = 4 dB². R is larger than the variance of the sequence after the moving average (0.1–2.2 dB²), so the Kalman filter trusts new samples little and smooths strongly. With this (Q, R) pair, the Kalman gain converges to $K \approx 0.083$, equivalent to an exponential filter with a time constant of about 12 samples (≈ 2.4 s for DeviceB, 3.0 s for DeviceA, 6.0 s for DeviceC); the −3 dB cutoff frequency of the Kalman filter is 0.055 Hz (DeviceA), 0.069 Hz (DeviceB), 0.028 Hz (DeviceC), and that of the N = 9 moving average is 0.20; 0.25; 0.10 Hz (Figure 2.5b). At these cutoff frequencies, most of the noise power has been removed.

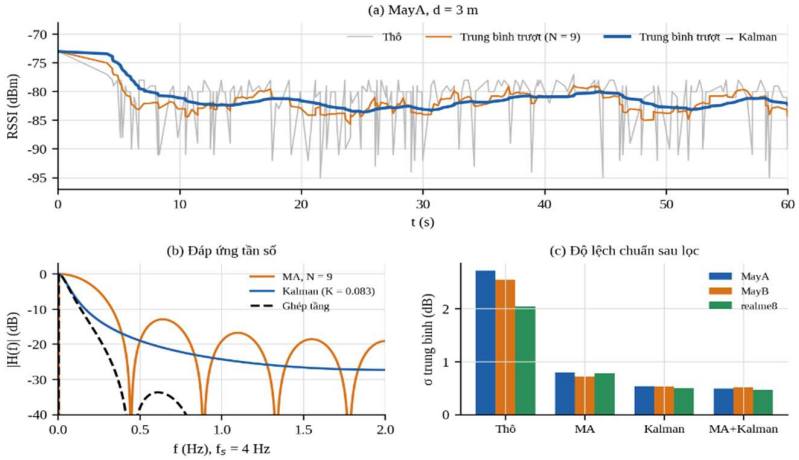


Figure 2.4: Filter performance: (a) DeviceA series at 3 m through each filter stage; (b) frequency response; (c) mean σ of RSSI by filtering step.

Results (Figure 2.5c): the mean σ decreases from 2.7 / 2.5 / 2.0 dB (raw) to 0.80 / 0.72 / 0.79 dB after the moving average and 0.49 / 0.52 / 0.47 dB (DeviceA / DeviceB / DeviceC) after both stages; the Kalman filter alone gives 0.54 / 0.53 / 0.51 dB, so the cascade is only slightly better. The price paid is a delay of a few seconds (Kalman time constant of 2–6 s plus the moving-average group delay of about 1–2 s) and a transient at the start of the sequence (Figure 2.2, DeviceA at the 3 m level), so it is suitable for devices that are *stationary* or *moving slowly*.

2.3. Real-time algorithm and evaluation methods

2.3.1. Distance formula and algorithm flowchart

From the filtered RSSI, the distance of each path is computed using the inverse of (2.1):

$$d = 10^{(Rssi_1m - Rssi)/(10\times n)} \quad (2.5)$$

With the origin at the PC, DeviceB at (W, 0), and DeviceA at (0, H), the three distances satisfy:

$d_A^2 = x^2 + (y - H)^2$, $d_B^2 = (x - W)^2 + y^2$, $d_C^2 = x^2 + y^2$.

Subtracting the equations pairwise yields x from (d_B, d_C) and y from (d_A, d_C):

$$x = (W^2 + d_C^2 - d_B^2) / (2W),\ y = (H^2 + d_C^2 - d_A^2) / (2H) \quad (2.6)$$

With W = 4 m and H = 4 m. Because there are three equations for two unknowns and all d_i are noisy, (2.6) does not use all the available information. The program takes (2.6) as the starting point and then finds the (x, y) that minimizes the weighted least-squares cost function:

$$J(x,\ y) = \sum_i w_i(\rho_i(x,\ y) - d_i)^2, \rho_i = \sqrt{(x - x_i)^2 + (y - y_i)^2} \quad (2.7)$$

where the anchors (x_i, y_i) are DeviceA (0, H), DeviceB (W, 0), and PC (0, 0) respectively, and the weights are $w_A = w_B = w_C = 1$ (the weight of a poorly calibrated path may be reduced). The problem is solved with Gauss–Newton with Levenberg–Marquardt damping at each iteration:

$$(J_r^T.W.J_r + \lambda.\mathrm{diag}(J_r^T.W.J_r)).s = -J_r^T.W.r$$

Where:

$r_i = \rho_i - d_i$ and the i-th row of the Jacobian J_r is $((x - x_i)/\rho_i, (y - y_i)/\rho_i)$.

A step s is accepted (and λ is reduced 10-fold) if J decreases; otherwise λ is increased 10-fold; the loop stops when $|s| < 0.01$ cm or after 30 iterations. If a path has had no new sample for more than 3 s, the program does not compute a position.

Position accuracy also depends on the geometry of the anchor layout. With independent distance errors of equal magnitude and equal weights, the position error is approximately GDOP × the distance error, where GDOP = $\sqrt{\mathrm{tr}((J_r^T J_r)^{-1})}$.

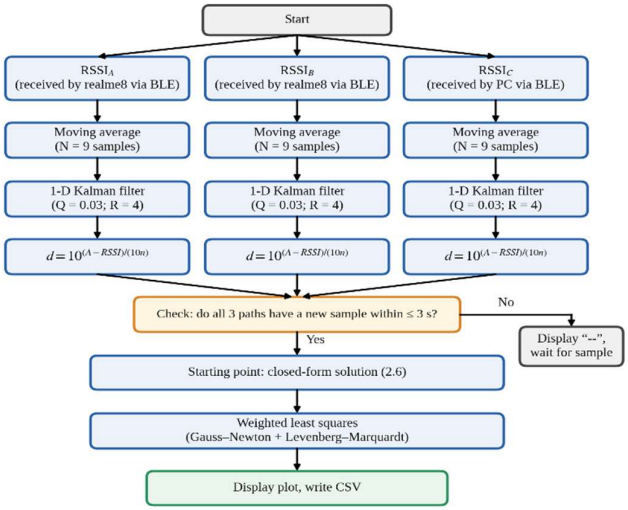


Figure 2.5: Flowchart of the real-time positioning algorithm.

2.3.2. Leave-one-out validation, RMSE, and the proportion of samples within ±0.5 m and ±1 m

Because samples within the same level are strongly correlated, leaving out individual samples would leak information; leave-one-out is therefore applied by distance level: one of the 5 levels is left out in turn, regression is performed on the remaining 4 levels (raw samples), and d is then estimated for every filtered sample (moving average → Kalman) of the left-out level; the estimated distance is limited to [0.1; 10] m. This is a demanding test because the 1 m and 5 m levels are extrapolations. The three metrics, with error $e = \hat{d} - d$, are: RMSE = $\sqrt{(\text{mean}(e^2))}$ (m); $P_{0.5}$ and P_1 are the proportions of samples with $|e| \leq 0.5$ m and $|e| \leq 1$ m.

Table 2.3: Leave-one-level-out results (RSSI filtered with MA + Kalman, 5 levels).

Path – model	RMSE (m)	$P_{0.5}$ (%)	P_1 (%)
DeviceA – log-distance	0.64	33.6	95.0
DeviceA – quadratic	0.74	20.8	90.8
DeviceB – log-distance	2.78	17.0	45.7
DeviceB – quadratic	1.10	19.2	66.5
DeviceC – log-distance	2.04	1.8	82.1
DeviceC – quadratic	0.64	58.9	91.8

Table 2.4: Leave-one-level-out RMSE (m) by filtering step.

Path (model)	Raw	MA	Kalman	MA + Kalman
DeviceA (log-distance)	1.92	0.76	0.65	0.64
DeviceB (quadratic)	1.18	1.10	1.10	1.10
DeviceC (quadratic)	0.76	0.61	0.63	0.64

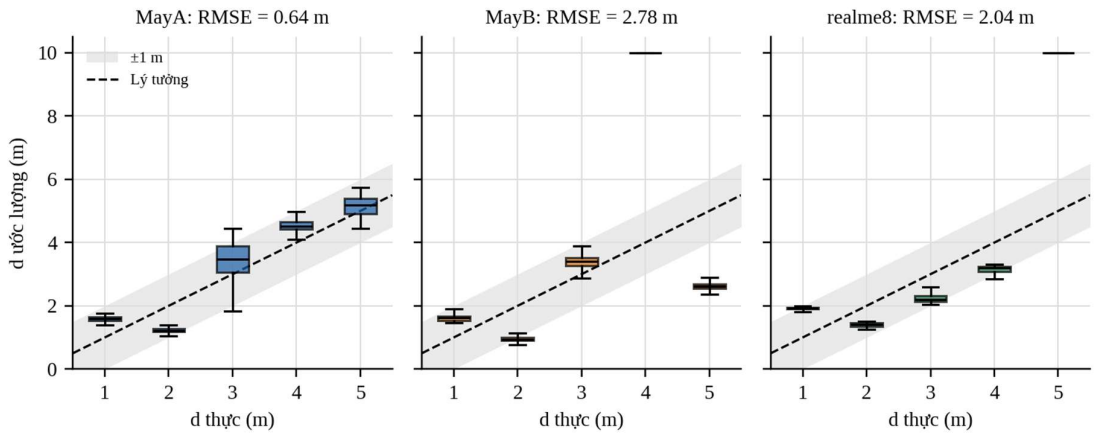


Figure 2.6: Estimated distance versus true distance (leave-one-level-out, log-distance model, filtered RSSI). Gray region: ±1 m error.

DeviceA with log-distance is the best path: RMSE 0.64 m, 95.0% of samples within ±1 m but only 33.6% within ±0.5 m. Filtering has a clear effect: RMSE decreases from 1.92 m (raw RSSI) to 0.64 m (Table 2.4), and the remaining error is mainly due to the model rather than noise.

The DeviceC path with the quadratic model gives an RMSE of 0.64 m and 91.8% of samples within ±1 m, whereas log-distance fails at 5 m (RMSE 2.04 m); the remaining error is concentrated at the 5 m level (RMSE 1.18 m, extrapolation).

DeviceB is the weakest path: the RSSI at 4 m (−88.3 dBm) is weaker than at 5 m (−82.7 dBm), so log-distance has an RMSE of 5.7 m at the 4 m level (overall RMSE 2.78 m, only 45.7% of samples within ±1 m). The quadratic model reduces the RMSE of DeviceB to 1.10 m (66.5% within ±1 m) but in effect merely follows the data and cannot be inverted at 5 m (0% within ±1 m).

➔ The non-monotonic RSSI of DeviceB and DeviceC in this experiment is caused by multipath reflections, antenna position, and especially device orientation.

2.3.3. Converting the ±1 m tolerance to dB at each distance

To see how demanding the ±1 m error requirement is for RSSI, the distance tolerance is converted to dB using (2.1):

$$\Delta RSSI\pm(d) = 10.n.|\log_{10}((d \pm 1)/d)| \quad (2.8)$$

For $d \geq 2$ m, the +1 m side is always tighter than the −1 m side ($\log_{10}$ increases slowly as d grows), so the tolerance is taken on the +1 m side; at d = 1 m only the +1 m side is meaningful. This tolerance is compared with the σ of the raw RSSI and the σ after filtering at each level (Table 2.5, Figure 2.9).

Table 2.5: +1 m tolerance converted to dB (Δ) and standard deviation of raw and filtered RSSI (dB) at each level.

d (m)	DeviceA			DeviceB			DeviceC		
	Δ (dB)	**σ raw**	**σ filt.**	**Δ (dB)**	**σ raw**	**σ filt.**	**Δ (dB)**	**σ raw**	**σ filt.**
1	4.87	1.18	0.45	3.85	3.29	0.43	7.31	1.29	0.29
2	2.85	2.41	0.37	2.25	3.43	0.72	4.28	1.94	0.36
3	2.02	4.88	0.89	1.60	1.25	0.32	3.03	2.05	0.44
4	1.57	2.00	0.32	1.24	3.46	0.84	2.35	2.42	0.53
5	1.28	3.26	0.44	1.01	1.67	0.31	1.92	2.72	0.72

The tolerance narrows quickly with distance: from 4.9 dB at 1 m to 1.3 dB at 5 m for DeviceA and 1.0 dB for DeviceB, i.e., only about one RSSI quantization step. With raw RSSI, σ exceeds the tolerance in 8 cases (DeviceA 3 m; DeviceA 4 m; DeviceA 5 m; DeviceB 2 m; DeviceB 4 m; DeviceB 5 m; DeviceC 4 m; DeviceC 5 m), so ±1 m is hard to achieve using raw samples.

After filtering, σ is at most 0.89 dB, lower than the tolerance at every level; however, the systematic model error (regression RMSE of 2.2–4.0 dB, Table 2.2) is still larger than the tolerance at 3–5 m. Therefore, ±1 m accuracy at larger distances is limited by how well the model fits, by the monotonicity of RSSI with distance, and by the differing hardware of each device, rather than by random noise.

CHAPTER 3. RESULTS

3.1. Survey and calibration results

RSSI data were collected continuously for 60 s at each mark from 1–5 m for 3 transmitting devices and 1 receiving device.

The statistical results show that the mean signal power decreases with measurement distance, and the standard deviation (Std) ranges from 1.18–4.88 dB, indicating multipath environmental noise. Power spectral density analysis confirms that the RSSI noise is wideband white noise in nature → entirely suitable for using low-pass filters.

Linear regression was performed over the reliable distance range of 1–4 m: the regression model achieves a high correlation (R^2=0.775-0.899) → the 95% confidence band widens toward both ends of the measurement range due to the edge effect of linear regression

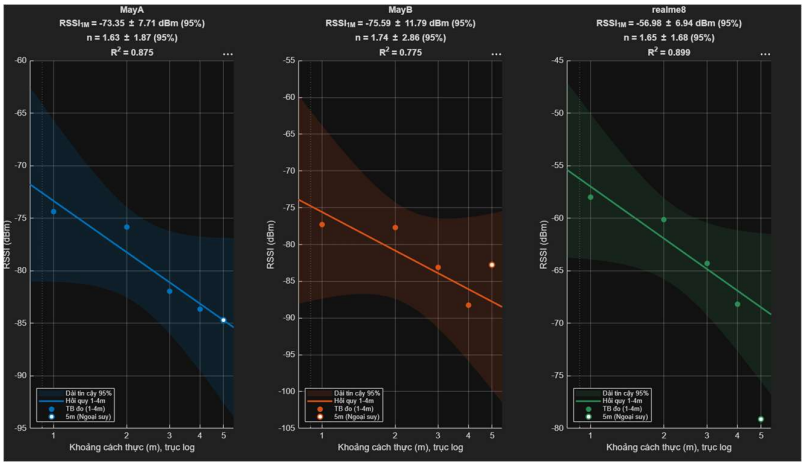
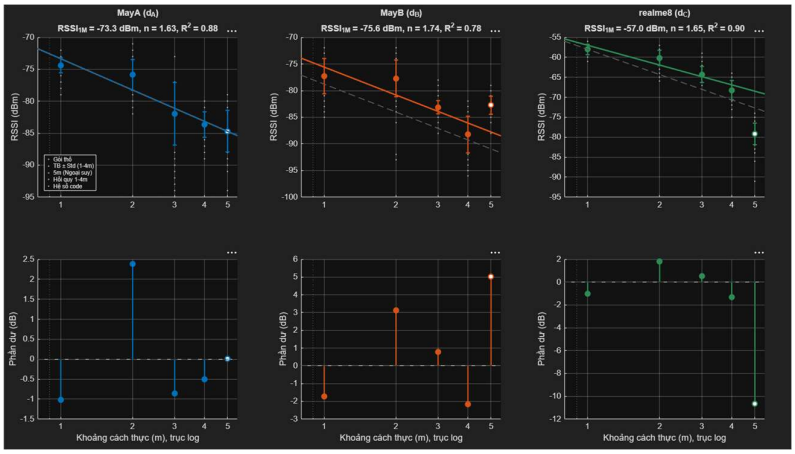


Figure 3.1: The coefficients RSSI_1M, n, R² and confidence intervals

In the 1–4 m range: the residuals of the points (by level mean) are randomly distributed around the 0 dB baseline → the linear model is not biased. However, at the 5 m mark there is an abnormally large deviation for DeviceB and realme8 → the 5 m mark was separated out as an extrapolation point and was not included in the calculation of the regression coefficients.



3.2. Noise filtering results

A moving average (MA) filter with a window of N=9 samples is used as preprocessing to remove spikes and bring the noise to a standard form → smoothing the BLE RSSI signal, with the aim of reducing fluctuation and measurement noise. Under the assumption of independent white noise, the output variance is reduced to 1/9 of the input variance, with a group delay of $\tau = \frac{N-1}{2} = 4$ samples, equivalent to 0.8–2 seconds at a sampling rate of 2–5 Hz.

Based on experimental data, the RSSI variance of Device A, Device B, and realme8. The Kalman filter is chosen with $Q=0.03$ and $R=4.0\ dB^2$, where:

Q denotes the process noise variance

R denotes the measurement noise variance.

With a scalar random-walk Kalman model with a ratio $\frac{Q}{R} = \frac{0.03}{4.0}$ yields a steady-state gain $K_{inf} \approx 0.083$, which reduces the influence of RSSI fluctuations and stabilizes the estimated value. Combining MA and Kalman helps improve signal stability to avoid distance jumps before computing the BLE distance in the Log-Distance model for the positioning algorithm.

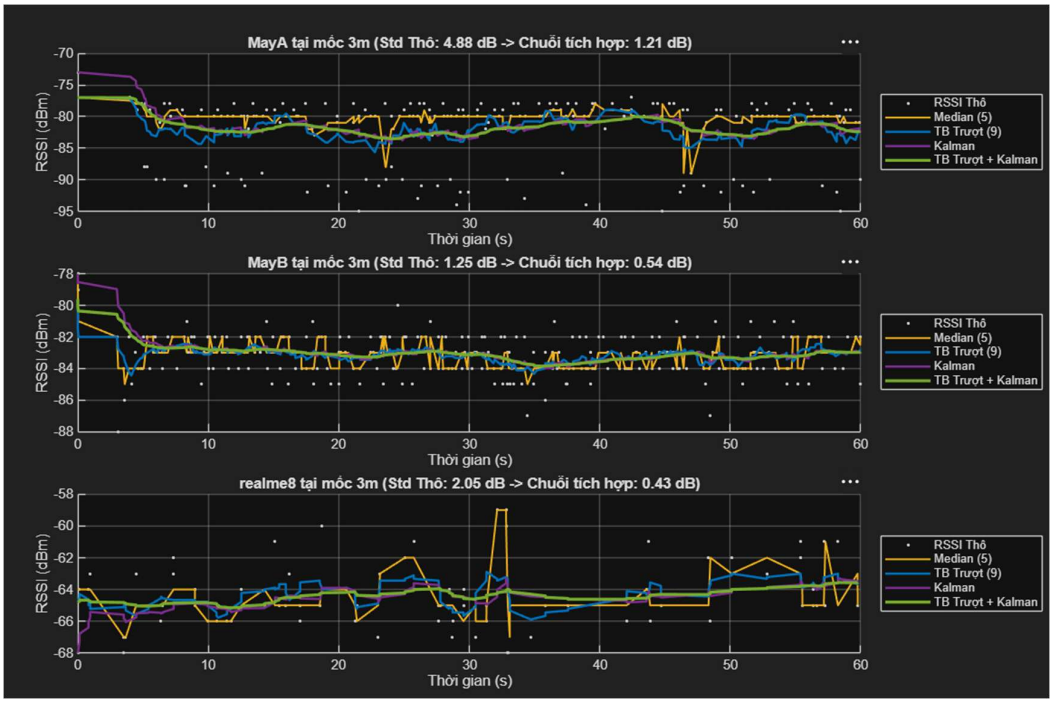


Figure 3.2: Comparison of raw, median, moving-average, and Kalman RSSI

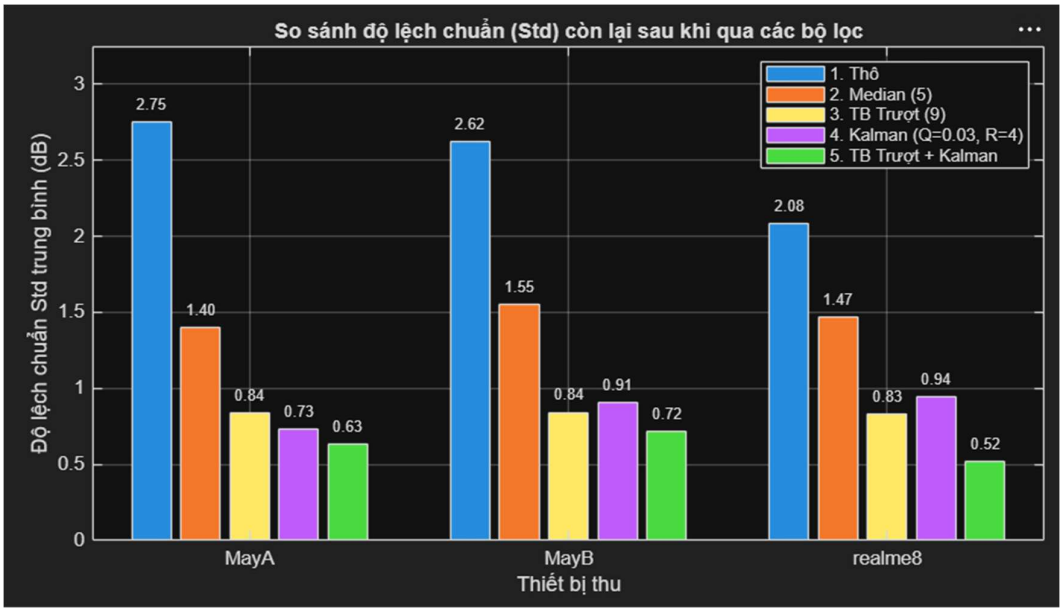


Figure 3.3: Comparison of the standard deviation (std) of RSSI across filters

When the moving average is combined with Kalman, the standard deviation drops to 0.63 dB for Device A, 0.72 dB for Device B, and 0.52 dB for realme8. Overall, combining the two filters yields the most stable signal among the configurations compared; however, it is still necessary to consider the response delay to ensure effectiveness when the device is moving.

3.3. Accuracy evaluation and conclusions

In real-world experiments, using the following parameters:

$$DeviceA: RSSI_{1M} = -73.35\ dBm, n = 1.63$$
$$DeviceB: RSSI_{1M} = -78.79\ dBm, n = 1.74$$
$$realme8: RSSI_{1M} = -58.00\ dBm, n = 2.10$$

The device performs the calculations quite accurately in experiments to determine the position of realme8 within the region (3 m, 4 m); the tested values fall within the error margin ± 1m.

Conclusion: The Log-Distance model showed high correlation and reliability in the 1–4 m range: R^2=0.775-0.899. Combined with the integrated filter chain of moving average plus Kalman, most of the white noise was eliminated. Choosing the experimental parameters for Device B and realme8 helped optimize the error performance.

References

[1] T. S. Rappaport, Wireless Communications: Principles and Practice, Upper Saddle River, NJ: Prentice Hall, 2002.

Source code: https://github.com/truongthuanphat2k5-art/Project.git